Recurrence-Aware Thompson Sampling
A Simple Algorithm for Bandits with Recurring Regimes

Abstract

We study a decision problem in which, at every round, a learner picks one of K actions and receives a reward. The catch is that the world is not fixed: it switches, from time to time, between a small number of hidden "regimes," and the *same* regimes keep coming back. A good learner should (i) quickly figure out which regime is currently active, (ii) exploit it, (iii) notice when it changes, and (iv) *remember* regimes it has already learned so that a returning regime is recognized instead of relearned from scratch. We present a deliberately minimal algorithm, *Recurrence-Aware Thompson Sampling* (RA-TS), that does all four using only three familiar pieces: Thompson sampling to choose actions, a change detector to spot when a regime ends, and a small memory of past regimes to warm-start on their return. This note explains the problem, the algorithm, and its guarantees in self-contained terms, states the algorithm in one pseudocode block, and lists every parameter in one table.

1 The problem, in plain terms

There are K *actions* (also called arms), $\mathcal{A} = \{1, \ldots, K\}$. Time proceeds in rounds $t = 1, 2, \ldots, T$. In round t the learner chooses an action $A_t \in \mathcal{A}$ and observes a numerical reward Y_t. The learner wants the total reward over T rounds to be as large as possible.

The environment is *piecewise stationary*: the timeline is cut into *episodes*, and inside one episode the rewards are governed by a single hidden *regime* z. When an episode ends, the regime changes. The key feature of our setting is *recurrence*: only a few distinct regimes ever occur, and they reappear over and over.

Level versus shape. Within a regime z, action k has an (unknown) average reward

$$\mu_k^{(z)} \;=\; \underbrace{\ell^{(z)}}_{\text{level}} + \underbrace{\theta_k^{(z)}}_{\text{shape}}, \qquad \sum_{k=1}^{K} \theta_k^{(z)} = 0. \tag{1}$$

The *level* $\ell^{(z)}$ is a number added to every action equally (an overall "how good are things right now" offset). The *shape* $\boldsymbol{\theta}^{(z)}$ says which actions are better or worse than average; it sums to zero by definition. The best action of a regime, $k^\star(z) = \arg\max_k \theta_k^{(z)}$, depends only on the shape. This distinction matters: if every reward suddenly rises or falls by the same amount, the level changes but the best action does not, so that is *not* a real regime change for our purposes. A real regime change is a change of *shape*.

To compare shapes we use *centering*. For any vector $\boldsymbol{v} \in \mathbb{R}^K$, subtract its own average:

$$\phi(\boldsymbol{v}) \;=\; \boldsymbol{v} - \frac{1}{K}\sum_{j=1}^{K} v_j. \tag{2}$$

Centering removes the level automatically: $\phi(\boldsymbol{\mu}^{(z)}) = \boldsymbol{\theta}^{(z)}$. Equivalently, we can compare actions by their *differences* $\mu_i - \mu_j$, which also do not depend on the level. This is why the algorithm never needs to estimate ℓ.

Noise. Observed rewards are the true mean plus zero-mean noise, $Y_t = \mu^{(z)}_{A_t} + \varepsilon_t$, where ε_t is σ-sub-Gaussian (informally: noise of typical size σ, with light tails). This is the only probabilistic assumption needed to state the algorithm.

Goal. We measure performance by *regret*: the total reward we lose compared with an oracle that always plays the best action of the current regime. Section **??** bounds it.

2 The idea in one paragraph

RA-TS combines three standard tools. **(1) Thompson sampling** chooses actions during normal operation: it keeps a probability estimate of each action's mean and plays the one that looks best under a random draw, which automatically balances trying and exploiting and needs no tuning. **(2) A change detector** watches the reward of the action we are playing and raises an alarm when it drifts away from what the current regime predicts. **(3) A memory ("library")** of regimes we have already learned lets us *recognize* a returning regime from its shape and *resume* its learning instead of starting over. The only extra ingredient is a very short, bounded "probe" that tries each action a few times right after a change, so that we have enough information to recognize the regime. Everything else is plain Thompson sampling.

3 What the algorithm remembers

The library. The library $\mathcal{L} = \{1, \ldots, J\}$ holds one entry per regime discovered so far. Entry g stores, for each action k, how many times that action has been played in regime g and the sum of the rewards seen:

$$N_{g,k} \text{ (count)}, \qquad S_{g,k} \text{ (reward sum)}, \qquad \widehat{\mu}_{g,k} = \frac{S_{g,k}}{N_{g,k}}, \qquad \widehat{\boldsymbol{\theta}}_g = \phi(\widehat{\boldsymbol{\mu}}_g). \tag{3}$$

The counts and sums (N_g, S_g) are the *only* things stored. The mean $\widehat{\boldsymbol{\mu}}_g$ and the shape $\widehat{\boldsymbol{\theta}}_g$ are *derived on demand* from them, never maintained separately: whenever they are read they simply reflect the current counts.

When (N_g, S_g)—and hence $\widehat{\boldsymbol{\theta}}_g$—change. This happens at exactly two kinds of event, and nowhere else:

(i) *Once, when a probe is assigned to regime g*: the whole identification buffer is added in one shot, $N_g \mathrel{+}= \boldsymbol{n}$, $S_g \mathrel{+}= \boldsymbol{s}$.

(ii) *Every* Track *round in regime g*: only the single action a just played is incremented, $N_{g,a} \mathrel{+}= 1$, $S_{g,a} \mathrel{+}= Y$.

So the shape estimate $\widehat{\boldsymbol{\theta}}_g$ is refreshed on *every* Track round (through $\widehat{\boldsymbol{\mu}}_g$), yet it is *read* in only one place: during Acquire, when a freshly collected probe buffer is matched against the library. The mean vector $\widehat{\boldsymbol{\mu}}_g$ is read once more, at the single moment a regime is identified and Track begins, to initialize the drift detector's baseline base $\leftarrow \widehat{\boldsymbol{\mu}}_g$; that value is then frozen for the entire Track phase and does not follow the subsequent updates to (N_g, S_g). Two consequences are worth noting. First, because Thompson sampling plays mostly the best action, Track sharpens mainly that one coordinate of $\widehat{\boldsymbol{\theta}}_g$; the other coordinates keep the coarser accuracy they had from the probe. Second, this causes no bias: each coordinate $\widehat{\mu}_{g,k} = S_{g,k}/N_{g,k}$ is an unbiased estimate of action k's mean no matter how few or how many times k was played, so $\widehat{\boldsymbol{\theta}}_g$ remains an unbiased shape estimate despite the very unequal counts. Its usefulness for matching is therefore limited by the *least*-sampled actions, i.e. by the radius $b(n_{\min})$—exactly the quantity that appears in the analysis. These raw statistics are also precisely the sufficient statistics Thompson sampling needs, so a regime is resumed just by continuing to add to them.

The scratch buffer and the baseline. Two more short-lived quantities are kept for the current situation: a *buffer* $(\boldsymbol{n}, \boldsymbol{s})$ of counts and sums collected while identifying the regime, and, once a regime is being tracked, a frozen *baseline* vector base $= \widehat{\boldsymbol{\mu}}_g$ (the regime's mean reward captured at the moment the regime is identified and TRACK begins) together with a per-action detector statistic G_k.

4 The two modes

Acquire (recognize the regime). Right after a change, we do not yet know which regime we are in. We run a short *coverage probe*: repeatedly play the least-tried action until every action has been tried $n_{\min}$ times. This guarantees we have a few samples of *every* action, which is exactly what is needed to estimate the shape. The moment coverage is reached, we compute the buffer's shape $\widehat{\boldsymbol{\psi}} = \phi(\boldsymbol{s}/\boldsymbol{n})$ and compare it to every library entry using the distance

$$D_g \;=\; \left\|\widehat{\boldsymbol{\psi}} - \widehat{\boldsymbol{\theta}}_g\right\|_\infty \qquad \text{(largest per-action shape difference).} \tag{4}$$

If the closest entry $g^\star$ is within a tolerance τ_{match}, we decide it is that regime and *warm-start*: we fold the buffer into entry $g^\star$ and continue its learning. Otherwise the shape matches nothing we know, so we create a new library entry. Either way we then switch to TRACK. Because comparison uses centered shapes (equivalently, action differences), it ignores the reward level entirely.

Track (exploit and watch). Now that the regime is known, we run Thompson sampling on its stored posterior. For each action we draw a random plausible mean and play the best draw:

$$\widetilde{\mu}_k \sim \mathcal{N}\big(m_{g,k},\, v_{g,k}\big), \qquad A_t = \arg\max_k \widetilde{\mu}_k, \tag{5}$$

where $(m_{g,k}, v_{g,k})$ is the usual Gaussian posterior mean and variance built from the prior $\mathcal{N}(\mu_0, \tau_0^2)$ and the stored data $(N_{g,k}, S_{g,k})$:

$$v_{g,k} = \left(\tfrac{1}{\tau_0^2} + \tfrac{N_{g,k}}{\sigma^2}\right)^{-1}, \qquad m_{g,k} = v_{g,k}\left(\tfrac{\mu_0}{\tau_0^2} + \tfrac{S_{g,k}}{\sigma^2}\right). \tag{6}$$

As more data accumulate, the draws concentrate and Thompson sampling plays the best action almost always: deterministic exploitation appears on its own, with no separate "commit" step.

At the same time, we watch for drift. After playing action a and seeing reward Y_t, we measure how surprising it is relative to the frozen baseline and accumulate the surprise in a one-sided CUSUM statistic:

$$e_t = Y_t - \text{base}[a], \qquad G_a \leftarrow \max\{0,\; G_a + |e_t| - \nu\}. \tag{7}$$

An alarm is raised when $G_a \geq h$, and we return to ACQUIRE. Three deliberate choices make this work: the baseline is *frozen* (a running average would silently follow the drift and hide it); the surprise uses the *absolute* residual $|e_t|$ (so both increases and decreases are caught); and the slack ν is set *above* the average noise, so with no drift the statistic is pushed back to zero (few false alarms), while a real shift makes it climb.

5 The algorithm and its parameters

Algorithm **??** states RA-TS in full. Table **??** lists every parameter, what it means, and how to set it.

Algorithm 1 Recurrence-Aware Thompson Sampling (RA-TS)

Input: actions K, noise scale σ, prior (μ_0, τ_0), coverage $n_{\min}$, match tolerance τ_{match}, detector slack ν, threshold h

Throughout: $\widehat{\boldsymbol{\mu}}_g = S_g/N_g$ *and* $\widehat{\boldsymbol{\theta}}_g = \phi(\widehat{\boldsymbol{\mu}}_g)$ *are recomputed from the current* (N_g, S_g).

library $\mathcal{L} \leftarrow \varnothing$; mode $\leftarrow$ Acquire; buffer $(\boldsymbol{n}, \boldsymbol{s}) \leftarrow (\mathbf{0}, \mathbf{0})$
for $t = 1, 2, \ldots, T$ **do**
  if mode = Acquire **then**
    play $a \leftarrow \arg\min_k n_k$, observe Y ▷ coverage probe: try the least-sampled action
    $n_a \leftarrow n_a + 1$; $s_a \leftarrow s_a + Y$
    if $\min_k n_k \geq n_{\min}$ **then** ▷ every action covered: identify the regime
      $\widehat{\boldsymbol{\psi}} \leftarrow \phi(\boldsymbol{s}/\boldsymbol{n})$; $g^\star \leftarrow \arg\min_{g \in \mathcal{L}} \left\| \widehat{\boldsymbol{\psi}} - \widehat{\boldsymbol{\theta}}_g \right\|_\infty$
      if $\mathcal{L} \neq \varnothing$ **and** $\left\| \widehat{\boldsymbol{\psi}} - \widehat{\boldsymbol{\theta}}_{g^\star} \right\|_\infty \leq \tau_{\text{match}}$ **then**
        $N_{g^\star} \leftarrow N_{g^\star} + \boldsymbol{n}$; $S_{g^\star} \leftarrow S_{g^\star} + \boldsymbol{s}$; $g \leftarrow g^\star$ ▷ match: warm-start
      else
        append new entry with $(N, S) \leftarrow (\boldsymbol{n}, \boldsymbol{s})$; $g \leftarrow$ new index ▷ new regime
      end if
      mode $\leftarrow$ Track; base $\leftarrow \widehat{\boldsymbol{\mu}}_g$; $\boldsymbol{G} \leftarrow \mathbf{0}$
    end if
  else ▷ mode = Track
    draw $\widetilde{\mu}_k \sim \mathcal{N}(m_{g,k}, v_{g,k})$ for all k ▷ Eq. (**??**)
    play $a \leftarrow \arg\max_k \widetilde{\mu}_k$, observe Y ▷ Thompson sampling
    $S_{g,a} \leftarrow S_{g,a} + Y$; $N_{g,a} \leftarrow N_{g,a} + 1$ ▷ resume/update posterior
    $G_a \leftarrow \max\{0,\ G_a + |Y - \text{base}[a]| - \nu\}$ ▷ drift CUSUM on played action
    if $G_a \geq h$ **then** ▷ alarm: regime ended
      mode $\leftarrow$ Acquire; buffer $(\boldsymbol{n}, \boldsymbol{s}) \leftarrow (\mathbf{0}, \mathbf{0})$
    end if
  end if
end for

6 Why each piece is there

Why a probe, and not just Thompson sampling? It is tempting to drop the probe and let Thompson sampling itself explore during Acquire. This does not work, for a simple reason. Recognizing a regime means telling the actions apart, which requires at least a few samples of *every* action, including the bad ones. But Thompson sampling is built to *stop* playing actions that look bad: after one or two samples a clearly inferior action is almost never chosen again. So coverage is never reached, identification never happens, the algorithm stays in Acquire, and—since the detector only runs in Track—a later regime change is not even watched. The probe supplies exactly the missing samples, and nothing more: at most $n_{\min}K$ plays, once per episode.

Why remember regimes? Without the library, every reappearance of a regime would be learned from scratch, paying the full exploration cost each time. With the library, a returning regime is recognized from its shape and its posterior is resumed, so its learning cost is paid essentially once and then amortized over all future visits. This is the main advantage under frequent switching among a fixed set of regimes.

Why watch only the played action? The detector updates G_a only for the action actually played, which is almost always the current best action. This is enough to catch any change that moves the best action's reward, and it costs nothing extra. The price is a blind spot: a change

Symbol	Meaning	How to set (and typical value)
K	number of actions	problem-given
σ	noise scale (sub-Gaussian proxy)	estimated from data
δ	failure probability of the confidence bounds	small, e.g. 0.1; enters $L = 2\log(8KT/\delta)$
T	horizon (number of rounds)	problem-given; used only inside L
μ_0	prior mean of the Thompson posterior	the average reward level; not sensitive
τ_0	prior standard deviation (diffuse)	large relative to reward spread, e.g. 0.5
$n_{\min}$	samples per action required by the probe	$n_{\min} > 32\,\sigma^2 L/\Delta_{\mathrm{sh}}^2$; small in practice (e.g. 3)
τ_{match}	shape-match tolerance	in $[\,b(n_{\min}),\ \Delta_{\mathrm{sh}} - b(n_{\min})\,]$; e.g. $0.8\,\Delta_{\mathrm{sh}}$
ν	detector slack (per-step "ignore" band)	$\sigma\sqrt{2/\pi} < \nu < \eta_{\min}$; e.g. $1.3\,\sigma$
h	detector alarm threshold	larger = fewer false alarms, slower detection; e.g. $8\,\sigma$
Derived (not tuned): $r(n) = \sigma\sqrt{L/n}$, $b(n) = 2\,r(n)$, $\phi(\boldsymbol{v}) = \boldsymbol{v} - \frac{1}{K}\sum_j v_j$.		

Table 1: All parameters of RA-TS. Δ_{sh} is the smallest shape separation between distinct regimes and $\eta_{\min}$ the smallest change of the played action's mean at a regime switch (Assumptions **??**–**??**); both are properties of the environment, not tuned. The radius $r(n)$ is the size of the estimation error after n samples; $b(n)$ is its centered (shape) counterpart.

that alters only actions we are *not* playing cannot be seen. This is unavoidable for any method that does not periodically re-test every action.

7 Guarantees

We now state what can be proved. We first list the assumptions in words, then give the regret bound and a sketch of why each part holds.

Assumption 1 (Separated regimes)**.** *Distinct regimes differ in shape by at least Δ_{sh}: for any two regimes that occur, $\left\|\boldsymbol{\theta}^{(z)} - \boldsymbol{\theta}^{(z')}\right\|_\infty \geq \Delta_{\mathrm{sh}} > 0$. (Regimes are actually distinguishable.)*

Assumption 2 (Visible changes)**.** *When a regime ends, the mean of the action we are committed to moves by at least $\eta_{\min}$. (A change shows up on the action we are actually playing.)*

Assumption 3 (Feasible detector and probe)**.** *The constants satisfy $\sigma\sqrt{2/\pi} < \nu < \eta_{\min}$ (the slack sits above the noise but below the visible change) and $n_{\min} > 32\,\sigma^2 L/\Delta_{\mathrm{sh}}^2$, i.e. $b(n_{\min}) < \Delta_{\mathrm{sh}}/4$ (a probe resolves the shape to finer than the regime gap), with $\tau_{\mathrm{match}} \in [\,b(n_{\min}),\ \Delta_{\mathrm{sh}} - b(n_{\min})\,]$.*

Theorem 1 (Regret bound, high probability)**.** *Let G_T be the number of episodes, $\mathcal{R}$ the set of distinct regimes, T_r the total number of* Track *rounds spent in regime r (so $\sum_r T_r \leq T$), $\Delta_{r,k}$ the reward gap of action k in regime r, and S_T the number of "short" episodes (shorter than the detection delay plus the probe). Under Assumptions* **??**–**??**, *with probability at least $1-\delta$,*

$$R_T \;\leq\; \underbrace{\sum_{r\in\mathcal{R}} \sum_{k\neq k^\star(r)} \frac{C\,\sigma^2 \log T_r}{\Delta_{r,k}}}_{(i)} \;+\; \underbrace{\Delta_{\max}\,(G_T + S_T)\Big(n_{\min}K + \tfrac{h}{\eta_{\min}-\nu}\Big)}_{(ii)} \;+\; \underbrace{\delta\,T\,\Delta_{\max}}_{(iii)},$$

for a universal constant C, where $\Delta_{\max}$ bounds the one-round regret.

Each term corresponds to one thing the algorithm does:

- **(i) Learning.** The cost of Thompson sampling finding the best action in each regime—paid *once per regime* and shared over all its recurrences (note the log of the cumulative lifetime T_r).
- **(ii) Identify + detect.** A fixed overhead per episode: the bounded probe ($\leq n_{\min}K$ plays) plus the detection delay ($\approx h/(\eta_{\min} - \nu)$ rounds). The S_T part is the small extra cost on "short" episodes—those shorter than this overhead, where a change can land inside the probe.
- **(iii) Failure.** The rare event that a confidence bound is violated.

Sketch of the learning term. All confidence estimates hold simultaneously on a "good event" of probability $\geq 1 - \delta$ (a union bound over actions and sample sizes, which is why $\log(8KT/\delta)$ appears in the radii). On that event, because a regime's posterior is only ever *added to* and never reset, all Thompson plays ever made in regime r behave like one uninterrupted Thompson run of length T_r on a fixed problem. The standard finite-time Thompson-sampling bound for such a run is $\sum_{k \neq k^\star} O(\sigma^2 \log T_r/\Delta_{r,k})$. The important point is that the logarithm is of the *cumulative* lifetime T_r, so revisiting a regime does not restart the exploration cost—this is the recurrence benefit.

Sketch of the probe and delay term. Each episode triggers at most one Acquire: a probe of at most $n_{\min}K$ forced plays, plus the rounds between the true change and the alarm. After a change the residual $|e_t|$ is at least $\eta_{\min}$ on average, so the CUSUM climbs at rate $\geq \eta_{\min} - \nu$ and crosses h in about $h/(\eta_{\min} - \nu)$ rounds. Each such round costs at most $\Delta_{\max}$.

Sketch of correct identification and the short-episode term. After a clean probe, the shape estimate is within $b(n_{\min})$ of the truth (Assumption **??**), which is closer than half the regime gap, so the nearest library entry is the right regime (if known) and otherwise a new entry is created—identification is correct. It can fail only when a change lands *inside* the short probe window, mixing two regimes; this happens only on episodes shorter than the probe-plus-delay overhead, and each such case costs at most one extra Acquire cycle. Finally, choosing $\nu > \sigma\sqrt{2/\pi}$ makes the detector's "no drift" increments negative on average, so false alarms are exponentially rare and contribute only to the δT failure budget.

Reading the bound. Only the first term depends on the gaps, and it is paid once per regime and then shared across all recurrences. The remaining terms are a small, fixed overhead per episode (identify + detect), with a little extra only for episodes too short to finish identifying. In the regime of interest—a few regimes recurring many times, each episode comfortably longer than the detection delay—the bound is dominated by the amortized Thompson term.